

AI511/MM505 Linear Algebra with Applications

Lecture 6 – Linear Transformations (continued), Compositions

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Outline

Recap

Euclidean vector spaces (continued)

Linear transformations (continued)

Examples of linear transformations

Compositions of linear transformations

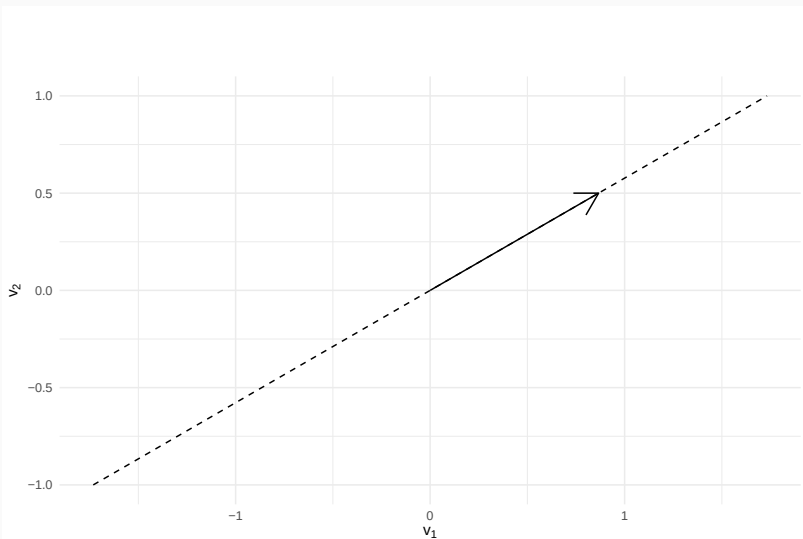
Recap

- When we say that v is a vector, we typically do not specify whether it is a column vector (i.e., $\mathbb{R}^{n \times 1}$) or a row vector (i.e., $\mathbb{R}^{1 \times n}$)
- When we think of vectors geometrically, this difference is often unimportant and we simply write

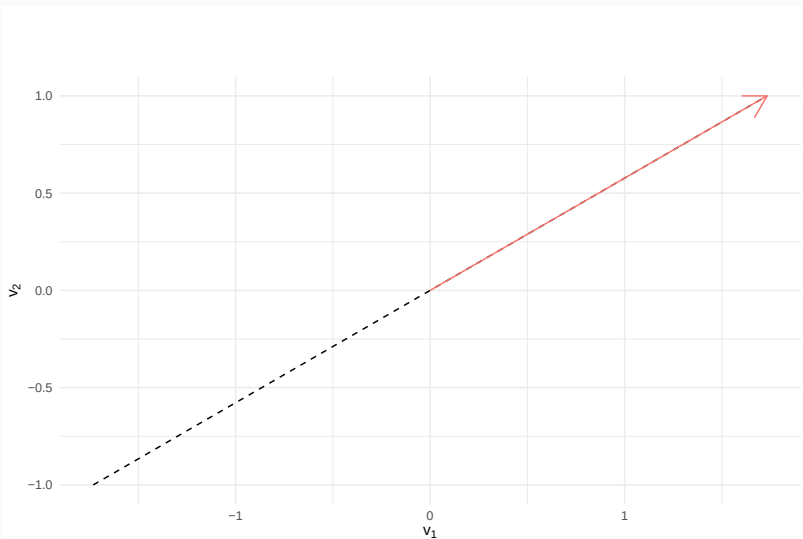
$$v = (v_1, \dots, v_n) \in \mathbb{R}^n.$$

- However, when we perform matrix multiplication, the difference between row vectors and column vectors is important.
- When the shape of a vector matters and we give no indication otherwise, vectors are assumed to be column vectors.

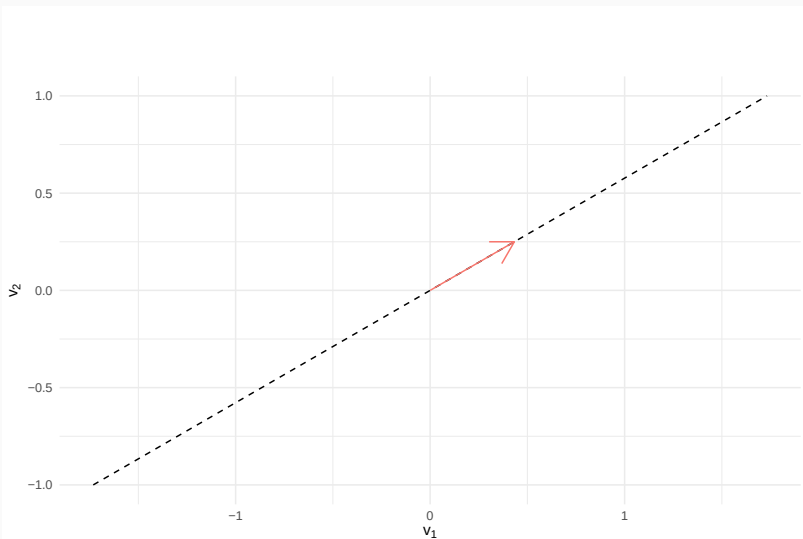
Scaling vectors in $\mathbb{R}^2$: $v = (\cos(\pi/6), \sin(\pi/6))$



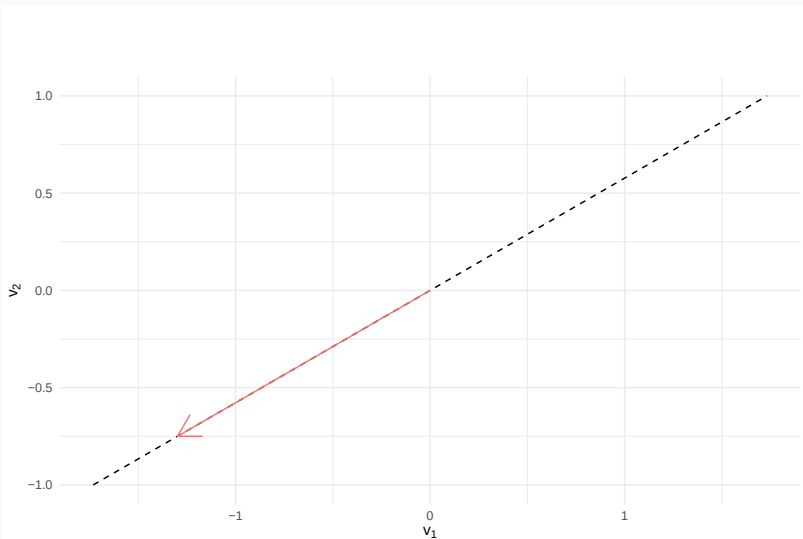
Scaling vectors in $\mathbb{R}^2$: $2v = 2(\cos(\pi/6), \sin(\pi/6))$



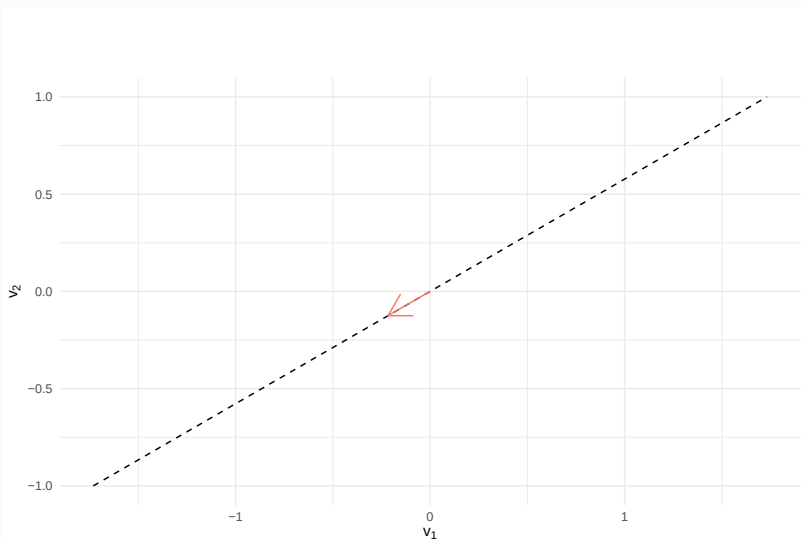
Scaling vectors in $\mathbb{R}^2$: $.5v = .5(\cos(\pi/6), \sin(\pi/6))$



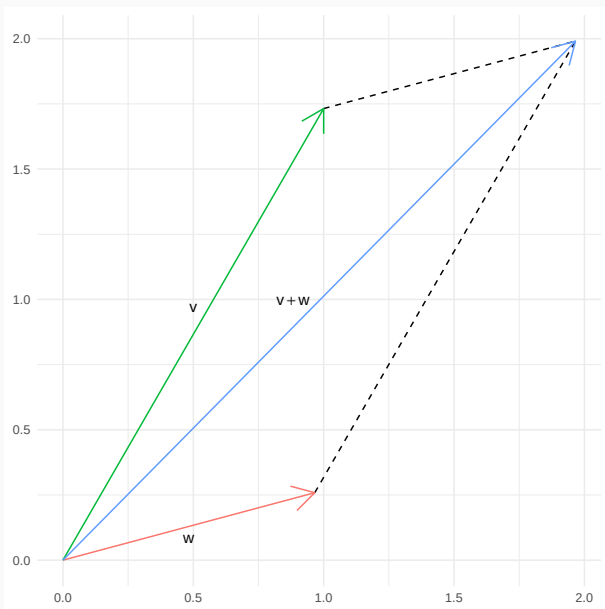
Scaling vectors in $\mathbb{R}^2$: $-1.5v = -1.5(\cos(\pi/6), \sin(\pi/6))$



Scaling vectors in $\mathbb{R}^2$: $-.25v = -.25(\cos(\pi/6), \sin(\pi/6))$



Vector addition in $\mathbb{R}^2$



Definition 1.2.1 of Johnston (2021)

Suppose $v = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$ and $w = (w_1, w_2, \dots, w_n) \in \mathbb{R}^n$. Then their *dot product*, denoted by $v \cdot w$, is the quantity

$$v \cdot w := v_1 w_1 + v_2 w_2 + \dots v_n w_n.$$

The dot product can be interpreted geometrically as roughly measuring the amount of overlap between v and w .

The angle between two vectors on the unit circle in $\mathbb{R}^2$

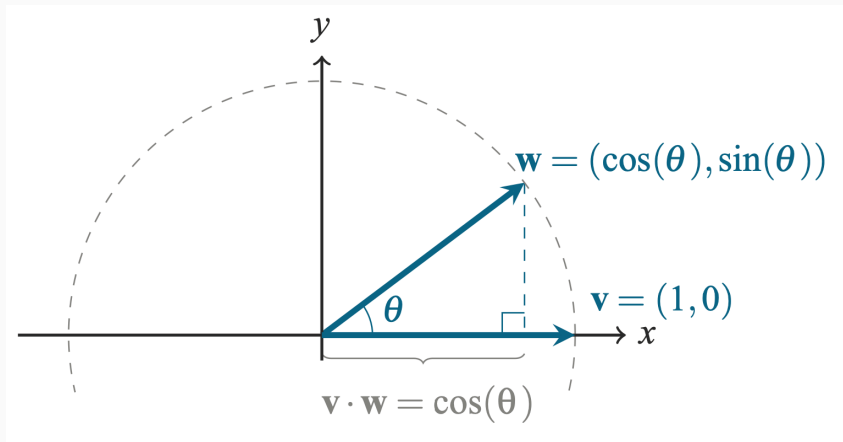


Figure 1.7 of Johnston (2021)

Length of a vector

Definition 1.2.2 of Johnston (2021)

The *length* of a vector $v = (v_1, v_2, \dots, v_n) \in \mathbb{R}^n$, denoted by $\|v\|$, is the quantity

$$\|v\| := \sqrt{v \cdot v} = \sqrt{v_1^2 + v_2^2 + \dots + v_n^2}.$$

$\|v\|$ is also called the Euclidean norm of the vector v .

Theorem 1.2.2 of Johnston (2021)

Suppose $v \in \mathbb{R}^n$ and $c \in \mathbb{R}$. Then the following properties hold:

- (a) $\|cv\| = |c|\|v\|$;
- (b) $\|v\| \geq 0$ with equality if and only if $v = 0$.

The standard basis

- The set of vectors $\{e_1, e_2, \dots, e_n\}$, where

$$e_1 = \begin{bmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}, \quad e_2 = \begin{bmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{bmatrix}, \quad \dots, \quad e_n = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 1 \end{bmatrix},$$

is called the *standard basis* in $\mathbb{R}^n$.

- Any vector $v \in \mathbb{R}^n$ can be written in terms of the vectors $e_1, e_2, \dots, e_n$ in the following way

$$v = v_1 e_1 + v_2 e_2 + \dots + v_n e_n. \tag{1}$$

Definition 1 on page 78 of Anton, Rorres, and Kaul (2019)

If T is a function with domain $\mathbb{R}^n$ and codomain $\mathbb{R}^m$, then we say that T is an *operator* or a *transformation* from $\mathbb{R}^n$ to $\mathbb{R}^m$ or that T *maps* from $\mathbb{R}^n$ to $\mathbb{R}^m$, which we denote by writing

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m.$$

Euclidean vector spaces (continued)

Euclidean vector spaces (continued)

Linear transformations (continued)

Matrices as transformations

- Consider now an $m \times n$ -matrix A .
- For every vector $x \in \mathbb{R}^n$, the product

$$Ax = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} a_{11}x_1 + \dots + a_{1n}x_n \\ a_{21}x_1 + \dots + a_{2n}x_n \\ \vdots \\ a_{m1}x_1 + \dots + a_{mn}x_n \end{bmatrix}$$

is a vector in $\mathbb{R}^m$.

- So we can think of A as a map $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by $T_A(x) := Ax$.
- T_A is called the *matrix operator* or *matrix transformation* associated with A .
- Recommended to watch *Linear transformations and matrices* of 3Blue1Brown.

Properties of matrix transformations

Theorem 1.8.1 of Anton, Rorres, and Kaul (2019)

For $A \in \mathbb{R}^{m \times n}$, $x, y \in \mathbb{R}^n$ and $k \in \mathbb{R}$, the following holds

- (1) $T_A(0) = 0$;
- (2) $T_A(k \cdot x) = k \cdot T_A(x)$ (homogeneity);
- (3) $T_A(x + y) = T_A(x) + T_A(y)$ (additivity);
- (4) $T_A(x - y) = T_A(x) - T_A(y)$.

- All these properties follow from our matrix multiplication rules.
Note that (2) and (3) together imply (1) and (4).

Linear transformation

Given some map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, how can we know if it is a matrix transformation? To answer this question, we need to learn what a linear transformation is.

Definition 1.4.1 of Johnston (2021)

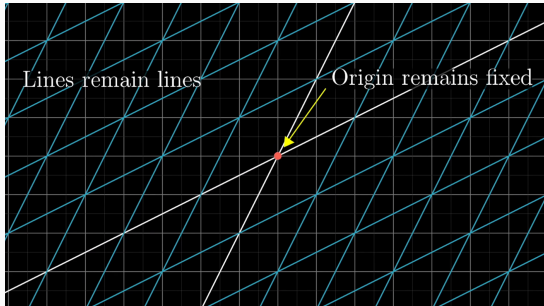
A *linear transformation* is a function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ that satisfies the following two properties:

- (a) $T(v + w) = T(v) + T(w)$ for all vectors $v, w \in \mathbb{R}^n$;
- (b) $T(cv) = cT(v)$ for all vectors $v \in \mathbb{R}^n$ and all scalars $c \in \mathbb{R}$.

Intuition

Visually speaking, a transformation is “linear” if it has two properties

- (i) all lines must remain lines, without getting curved;
- (ii) the origin must remain fixed in place.



Linear transformations and matrices by 3Blue1Brown

Standard matrix of a linear transformation

Theorem 1.4.1 of Johnston (2021)

A function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is a linear transformation if and only if there exists a matrix $[T] \in \mathbb{R}^{m \times n}$ such that

$$T(v) = [T]v \quad \text{for all } v \in \mathbb{R}^n.$$

Furthermore, the unique matrix $[T]$ with this property is called the *standard matrix* of T , and it is

$$[T] := \begin{bmatrix} T(e_1) & T(e_2) & \dots & T(e_n) \end{bmatrix}.$$

Observe that $[T]$ denotes a matrix.

Partial proof

Proof.

We prove that any linear transformation can be represented as $T(v) = [T]v$ for all $v \in \mathbb{R}^n$. Since any vector can be expressed using the standard basis and the assumption that T is linear,

$$\begin{aligned} T(v) &= T(v_1e_1 + v_2e_2 + \dots + v_ne_n) \\ &= v_1T(e_1) + v_2T(e_2) + \dots + v_nT(e_n) \\ &= \begin{bmatrix} T(e_1) & T(e_2) & \dots & T(e_n) \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \\ &= [T]v. \end{aligned}$$

Example on the blackboard

- To find the standard matrix $[T]$ of a linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, compute the column vectors $T(e_1), \dots, T(e_n) \in \mathbb{R}^m$ to form the columns of $[T]$, i.e.,

$$[T] = \begin{bmatrix} T(e_1) & \dots & T(e_n) \end{bmatrix}.$$

- Find the standard matrix of the following linear transformations
 - (a) $T(v_1, v_2) = (v_1 + 2v_2, 3v_1 + 4v_2)$;
 - (b) $T(v_1, v_2, v_3) = (3v_1 - v_2 + v_3, 2v_1 + 4v_2 - 2v_3)$.

Euclidean vector spaces (continued)

Examples of linear transformations

Simplest linear transformations

- The *zero transformation* $0 : \mathbb{R}^n \rightarrow \mathbb{R}^m$, defined by

$$0(v) = 0$$

for all $v \in \mathbb{R}^n$.

- The *identity transformation* $I : \mathbb{R}^n \rightarrow \mathbb{R}^n$, defined by

$$I(v) = v$$

for all $v \in \mathbb{R}^n$.

- What are the standard matrices of these transformations?

Section 1.4.2 of Johnston (2021)

Orthogonal projections onto the x - and y -axes

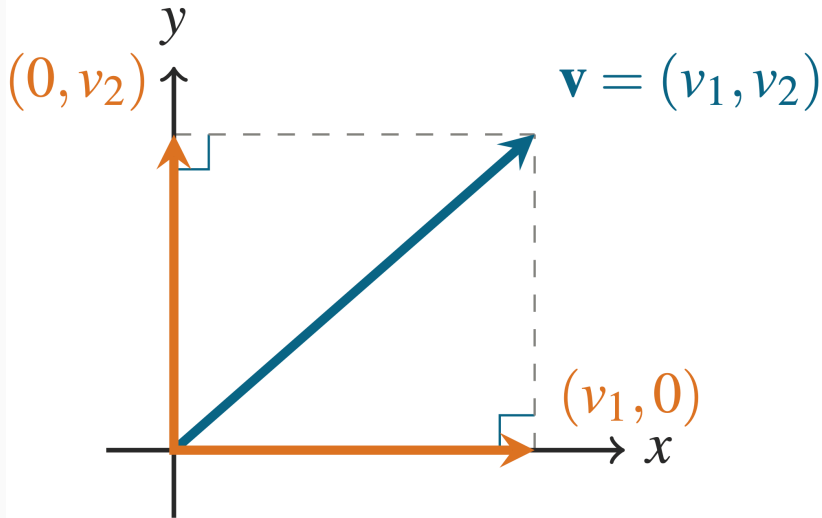


Figure 1.16 of Johnston (2021)

Orthogonal projection onto the direction of a unit vector u

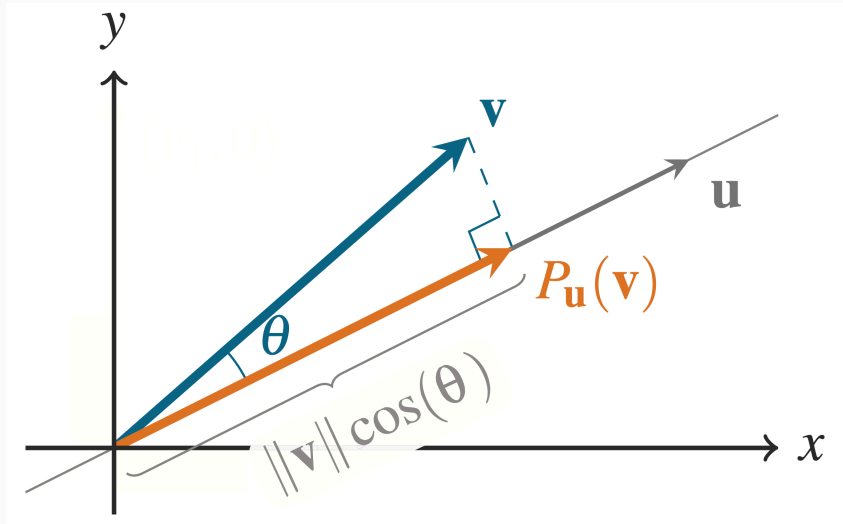


Figure 1.16 of Johnston (2021)

Orthogonal projection onto the direction of a unit vector u

- The projection of a vector v onto the line in the direction of u is denoted by $P_u(v)$, and it points a distance of $\|v\| \cos(\theta)$ (might be negative) in the direction of u , where θ is the angle between v and u .
- For $v, w \in \mathbb{R}^2$, we have that

$$\cos(\theta) = \frac{v \cdot w}{\|v\| \|w\|},$$

$$P_u(v) = \|v\| \cos(\theta) u = \|v\| \frac{(v \cdot u)}{\|v\| \|u\|} u = uu^T v$$

using the fact that $\|u\| = 1$ and $v \cdot u = u^T v$.

- Hence, $P_u(v)$ is a linear transformation, and its standard matrix is given by

$$[P_u] = uu^T \in \mathbb{R}^{n \times n}.$$

Orthogonal projection onto the direction of a vector w

In general, we have that

$$P_w(v) = \frac{1}{\|w\|^2} ww^T v = \frac{ww^T}{w^T w} v$$

for two vectors $v, w \in \mathbb{R}^n$ provided that $w \neq 0$.

Euclidean vector spaces (continued)

Compositions of linear transformations

Composition of linear transformations

Definition 1.4.2 of Johnston (2021)

Suppose $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $S : \mathbb{R}^m \rightarrow \mathbb{R}^p$ are linear transformations. Then *composition* of S and T is the function $S \circ T : \mathbb{R}^n \rightarrow \mathbb{R}^p$ defined by

$$(S \circ T)(v) = S(T(v)) \quad \text{for all } v \in \mathbb{R}^n.$$

- T sends $\mathbb{R}^n$ to $\mathbb{R}^m$ and S sends $\mathbb{R}^m$ to $\mathbb{R}^p$ while the composition $S \circ T$ skips the intermediate step and sends $\mathbb{R}^n$ directly to $\mathbb{R}^p$.
- Is $S \circ T$ linear? If so, what is its standard matrix?

Composition is linear

Theorem 1.4.2 of Johnston (2021)

Suppose $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $S : \mathbb{R}^m \rightarrow \mathbb{R}^p$ are linear transformations with standard matrices $[T] \in \mathbb{R}^{m \times n}$ and $[S] \in \mathbb{R}^{p \times m}$, respectively. Then $S \circ T : \mathbb{R}^n \rightarrow \mathbb{R}^p$ is a linear transformation, and its standard matrix is

$$[S \circ T] = [S][T].$$

- In simple terms, composition of linear maps is a matrix product of standard matrices.
- This theorem is the main reason that matrix multiplication is defined in the seemingly bizarre way.
- Recommended to watch *Matrix multiplication as composition* of 3Blue1Brown.

Example on the blackboard

- Consider two linear transformations

$$A = \begin{bmatrix} 0 & 2 \\ 1 & 0 \end{bmatrix} \quad \text{and} \quad B = \begin{bmatrix} 1 & -2 \\ 1 & 0 \end{bmatrix}.$$

- The linear transformation B
 - (i) rotates e_1 counterclockwise by 45° and extends its length to $\sqrt{2}$;
 - (ii) rotates e_2 counterclockwise by 90° and extends its length to 2.
- The linear transformation A
 - (i) flips e_1 and e_2 around 45° line;
 - (ii) extends the length of e_2 to 2 (stretches the space horizontally by 2).
- Find the standard matrix of the composition AB by describing where e_1 and e_2 are mapped and confirm your result by multiplying A and B .

References



Johnston, Nathaniel (2021). *Introduction to linear and matrix algebra*. Springer Cham.



Anton, Howard, Chris Rorres, and Anton Kaul (Sept. 2019). *Elementary linear algebra: applications version*. 12th edition. John Wiley & Sons, Inc.